

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Third Semester B.E. (RI) (Credit System) Degree Examinations**

April 2023

21RI301 - NUMERICAL METHODS AND TRANSFORM TECHNIQUES

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I**Marks BT* CO* PO***

1. a) Apply Lagrange's method to find
- $f(10)$
- .

x	5	6	9	11
f(x)	12	13	14	16

6 L*2 1 1

- b) Find the missing term in the following table.

x	1	2	3	4	5	6
y	2	6	12	-	30	42

7 L3 1 2

- c) Using Newton-Raphson method obtain the root near the given initial approximation of the root for the following (carry out 3 iterations).

$$\cos x = xe^x \text{ with } x_0 = 1$$

7 L1 2 1

2. a) Using fourth order Runge-Kutta method solve the following initial value problem
- $y' = 1 + y^2$
- ;
- $y(0) = 0$
- . Find
- $y(0.2)$
- taking
- $h = 0.2$

6 L2 1 1

- b) Obtain the natural cubic spline in the interval (1,2) for the following data.

x	1	2	3
y	-8	-1	18

Use the end condition $y_0'' = 0 = y_3''$

7 L2 2 1

- c) Using Newton-Raphson method solve the system of non-linear equations. (carry out two iterations)
- $x - 0.7 \sin x - 0.2 \cos y = 0$
- and
- $y - 0.7 \cos x + 0.2 \sin y = 0$
- given
- $x_0 = 1$
- &
- $y_0 = 1$
- .

7 L2 2 1

3. a) Using Bisection method obtain the root of the equation in the interval (2.5,3),
- $x^3 - 4x - 9 = 0$
- . Carry out five iterations.

6 L2 2 1

- b) Using Modified Euler's method find an approximate value of y when
- $x = 0.2$
- , taking
- $h = 0.1$
- . Given
- $y' = \frac{y-x}{y+x}$
- ;
- $y(0) = 1$

7 L1 2 1

- c) Find the cubic polynomial which takes the following values

x	0	1	2	3
f(x)	1	2	1	10

Hence evaluate $f(4)$.

7 L2 1 2

Unit – II

4. a) Find the Fourier series of for the function
- $f(x) = x - x^2$
- in the interval
- $-1 < x < 1$
- .

6 L2 3 1

P.T.O.

- b) Find the Fourier Transform of

$$f(x) = \begin{cases} 1 & ; |x| \leq 1 \\ 0 & ; |x| > 1 \end{cases}$$

Hence evaluate $\int_0^{\infty} \frac{\sin x}{x} dx$.

7 L1 3 1

- c) Find the Z-transform and ROC of
- $u_n = \begin{cases} 3^{-n}, & \text{if } n \geq 0 \\ 1, & \text{if } n < 0 \end{cases}$

7 L1 4 1

5. a) Find the Half range Fourier cosine series of
- $f(x) = x$
- in
- $(0, 2)$

6 L2 3 2

- b) Find
- $Z(3u_n - 2v_n)$
- where
- $u_n = \begin{cases} 1, & \text{if } n \geq 0 \\ 3^n, & \text{if } n < 0 \end{cases}$
- and
- $v_n = \begin{cases} 2^n, & \text{if } n \geq 0 \\ 5^n, & \text{if } n < 0 \end{cases}$

7 L2 4 2

- c) State and prove initial and final value theorem of unilateral Z-transforms.

7 L1 4 1

6. a) Find the Fourier series of the function
- $f(x) = x^2$
- in
- $(-\pi, \pi)$
- .

6 L1 3 1

- b) Find the Fourier sine transform of
- $\frac{e^{-ax}}{x}$
- .

7 L2 3 1

- c) State and prove Convolution theorem for Bilateral Z-Transform.

7 L1 4 1

Unit – III

7. a) With the usual notation prove that
- $\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0$
- .

6 L1 5 1

- b) Find the maximum value of
- $z = 2x + 3y$
- subject to the constraints
- $x + y \leq 30$
- ;
- $y \geq 3$
- ;
- $0 \leq y \leq 12$
- ;
- $x - y \geq 0$
- and
- $0 \leq x \leq 20$
- . Solve the above LPP by graphical method.

7 L2 5 1

- c) Prove that a heavy cable hangs freely under gravity, between two fixed points, then the shape of the cable is a catenary.

7 L2 5 2

8. a) A firm manufactures 3 products A, B and C. The profits are Rs 3, Rs 2 and Rs 4 respectively. The firm has two machines M1 and M2 and below is the required processing time in minutes for each machine on each product.

Machine on each product:				
Machine	Product			
		A	B	C
	M1	4	3	5
	M2	2	2	4

Machines M1 and M2 have 2000 and 2500 machine - minutes respectively. The firm must manufacture 100 A's, 200 B's and 50 C's but not more than 150 A's. Set up an LPP to maximize the profit.

6 L2 5 1

- b) Find the curve on which the functional
- $\int_0^1 [x + y + (y')^2] dx = 0$
- with
- $y(0) = 1$
- &
- $y(1) = 2$
- can be extremized.

7 L2 5 1

- c) Solve the following LPP by simplex method.

$$\text{Max } z = 107x_1 + x_2 + 2x_3$$

$$\text{Sub to } \frac{14}{3}x_1 + \frac{1}{3}x_2 - \frac{6}{3}x_3 \leq \frac{7}{3}$$

$$16x_1 + \frac{1}{2}x_2 - 6x_3 \leq 5$$

$$3x_1 - x_2 - x_3 \leq 0$$

$$\text{And } x_1, x_2, x_3 \geq 0$$

7 L2 5 1

BT* Bloom's Taxonomy, L* Leve CO* Course Outcome; PO* Program Outcome.
